

CFB GAGETOWN, NB, Canada

WMO: 717010

Lat: 45.8384N

Lon: 66.4331W

Elev: 51

StdP: 100.72

Time Zone: -4.00 (NAA)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -22.0	(c) -19.4	(d) -29.3	(e) 0.3	(f) -20.8	(g) -26.3	(h) 0.3	(i) -18.6	(j) 11.1	(k) -7.2	(l) 10.1	(m) -6.9	(n) 3.4	(o) 280	(p) 0.587

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	29.3	21.3	27.6	20.2	26.1	19.3	22.6	27.6	21.5	25.9	20.5	24.3	4.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	15.6	25.1	20.0	14.8	24.2	19.1	13.9	22.7	66.8	27.5	63.0	26.0	59.3	24.4	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.5	(b) 8.4	(c) 7.4	(e) -25.9	(f) 32.7	(g) 3.0	(h) 1.3	(i) -28.1	(j) 33.6	(k) -29.8	(l) 34.4	(m) -31.5	(n) 35.1	(o) -33.6	(p) 36.0		
(a) 9.5	(b) 8.4	(c) 7.4	(e) -26.3	(f) 24.4	(g) 2.9	(h) 0.8	(i) -28.4	(j) 25.0	(k) -30.0	(l) 25.5	(m) -31.7	(n) 26.0	(o) -33.7	(p) 26.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-8.2	-7.0	-1.8	4.9	11.4	16.2	19.8	19.0	14.9	8.5	2.4	-4.3
	DBStd	10.72	6.48	5.93	5.27	4.03	3.74	3.51	2.77	2.87	4.05	4.26	4.88	5.67
	HDD10.0	2354	566	475	367	158	26	1	0	0	6	81	231	443
	HDD18.3	4503	823	708	624	402	218	82	16	26	117	307	477	702
	CDD10.0	1037	0	0	1	6	68	188	304	280	153	33	4	0
	CDD18.3	144	0	0	0	0	3	18	62	47	14	1	0	0
	CDH23.3	1146	0	0	2	4	56	182	474	328	97	4	0	0
	CDH26.7	254	0	0	0	1	12	43	109	64	24	0	0	0
	WSAvg	3.4	3.7	3.9	4.0	3.8	3.5	3.2	2.9	2.8	3.0	3.3	3.6	3.7
	PrecAvg	1151	105	87	96	86	96	85	88	91	95	97	111	116
(15) Precipitation	PrecMax	1520	223	164	194	155	211	199	174	238	271	182	206	206
	PrecMin	845	20	28	27	20	20	26	29	17	20	21	44	48
	PrecStd	162	43	35	36	34	47	41	35	51	45	43	39	46
	PrecStd	162	43	35	36	34	47	41	35	51	45	43	39	46
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	9.7	15.6	21.4	28.1	30.5	31.5	30.8	29.8	23.7	17.4	12.6
		MCWB	11.0	8.6	8.8	11.7	17.3	21.1	22.6	22.5	21.9	16.5	15.8	11.7
	2%	DB	7.1	5.9	9.7	17.3	24.4	27.6	29.3	28.4	26.0	19.7	14.7	9.0
		MCWB	6.3	4.1	5.6	10.2	15.7	19.3	21.6	21.1	19.8	15.3	12.8	8.3
	5%	DB	3.6	3.7	7.5	14.5	21.4	25.4	27.7	26.7	23.5	17.5	12.5	6.1
		MCWB	2.3	1.9	4.7	9.3	13.8	18.1	20.6	20.0	18.3	14.4	10.6	4.9
	10%	DB	1.0	1.6	5.4	12.1	18.9	23.2	26.1	25.1	21.6	15.7	9.9	3.6
		MCWB	0.0	0.1	2.5	7.8	12.4	16.8	19.7	19.1	17.3	12.8	8.2	2.4
	0.4%	WB	11.1	8.9	11.0	14.9	19.5	22.8	24.0	23.5	22.6	18.5	16.1	12.0
		MCDB	11.7	9.4	13.3	18.1	26.6	28.8	29.2	28.8	28.3	21.4	17.1	12.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	6.5	4.7	7.1	12.5	17.0	20.8	22.7	22.3	20.8	16.6	13.5	8.3
		MCDB	6.9	5.7	8.4	15.0	22.0	25.4	27.8	26.8	24.2	18.8	14.5	9.0
	5%	WB	2.4	2.1	4.7	10.2	15.2	19.3	21.7	21.2	19.5	15.0	11.0	5.3
		MCDB	3.4	3.5	7.0	13.5	19.4	23.5	26.2	25.1	22.3	17.3	12.1	5.9
	10%	WB	0.1	0.3	2.8	8.1	13.5	17.9	20.5	20.2	18.1	13.1	8.1	2.5
		MCDB	0.9	1.5	5.1	11.6	17.7	21.9	24.6	23.5	20.6	15.1	9.5	3.3
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.6	9.4	10.0	11.5	11.1	11.0	10.9	11.3	9.5	7.7	7.8
		MCDBR	11.8	10.8	11.8	14.9	16.3	14.6	13.1	13.0	12.7	12.1	10.6	10.2
		MCWBR	11.8	9.5	8.9	9.4	9.0	7.9	6.7	6.8	7.5	8.5	9.4	10.3
	5% WB	MCDBR	11.8	10.3	10.5	12.4	13.8	12.2	11.4	10.9	10.9	10.7	10.0	10.6
		MCWBR	11.9	9.5	8.9	9.4	8.8	7.4	6.4	6.4	7.2	8.5	9.8	10.9
(40) Clear-Sky Solar Irradiance	taub		0.276	0.285	0.316	0.348	0.372	0.406	0.425	0.398	0.367	0.327	0.305	0.286
	taud		2.451	2.414	2.442	2.420	2.395	2.304	2.271	2.348	2.410	2.526	2.540	2.476
	Ebn at Noon		842	902	917	912	896	862	839	850	848	842	802	787
	Edh at Noon		68	85	97	108	115	127	130	115	100	77	63	60
	RadAvg		1.50	2.36	3.51	4.34	4.99	5.39	5.49	4.86	3.93	2.44	1.43	1.11
(45) All-Sky Solar Radiation	RadStd		0.15	0.21	0.23	0.41	0.57	0.42	0.43	0.31	0.29	0.23	0.16	0.10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	+52	N/A

Nomenclature: See separate page